

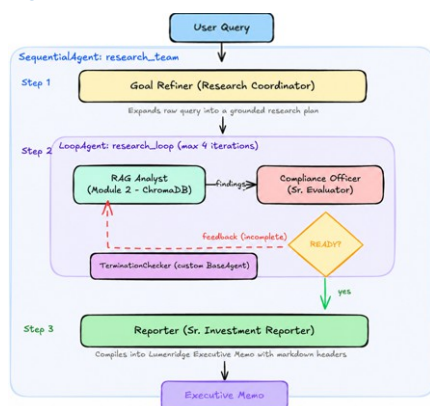
# Enterprise Agentic Workflows & Multi-Agent Systems

## DeepFin Investor Workshop summary with Google

In this research note, we summarise our September 2026 DeepFin Tutorial, “Enterprise Agentic Workflows & Multi-Agent Systems for Investment Strategy”, delivered with Google. We explore single-purpose agents, to manager delegation, and onto an autonomous research team. The key message is that investment workflows need governed autonomy, with review and traceability designed in from the start.

- Agentic AI: capabilities and governed autonomy**  
 Agentic AI shifts the focus from models that answer questions to systems that can plan, act, and reflect toward an objective. For investment workflows, the key is not autonomy alone, but bounded autonomy: permissioned tools, approved data sources, review points, and escalation paths. The opportunity is not simply to make research faster, but to automate repeatable research steps while keeping outputs traceable and fit for decision-making.
- Specialist Agents with live data and grounded RAG capabilities**  
 The first modules built focused agents with clear mandates. One assistant used live market-data tools to retrieve current inputs and run calculations at query time, while a second used RAG to ground answers in approved private documents. Together, they show how tool access turns a generic model into a workflow-specific research assistant, provided retrieval quality, missing-data behaviour, and outputs are tested carefully.
- Assembling the components to build an Autonomous Research Team**  
 Multi-agent coordination was introduced through manager-led delegation, where a routing agent interprets intent, assigns work to specialist agents, and consolidates outputs. Finally, these components were combined into an end-to-end research workflow with explicit roles including a goal refiner, a three-agent iterative research loop with compliance review and termination checks, and a reporter that produces a final memo once acceptance criteria are met. We find this is the more *practical* enterprise model: not an unconstrained agent, but a governed research harness with explicit roles, limits, and evidence standards.

Figure 1: Autonomous Financial Research Team



Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

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This workshop was run in conjunction with Google. For more details on the Agent Development Toolkit please see <https://adk.dev/get-started/about/>



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## Agentic AI - A Brief Introduction

Our recent DeepFin Investor Tutorial, delivered in partnership with Google, entitled **Enterprise Agentic Workflows & Multi-Agent Systems for Investment Strategy** focussed on equipping investors with the knowledge, tools and practical hands-on experience of composing, building and deploying agents. Thinking about the journey from automation to autonomous intelligence, workflows are systems where LLMs and tools are orchestrated through predefined code paths. Agents, on the other hand, are systems where LLMs dynamically direct their own processes and tool usage, maintaining control over how they accomplish tasks.

“Plan” refers to decomposing an objective into a sequence of steps.

“Act” refers to the use of tools to interact with the world, such as retrieving data, running code, or calling APIs.

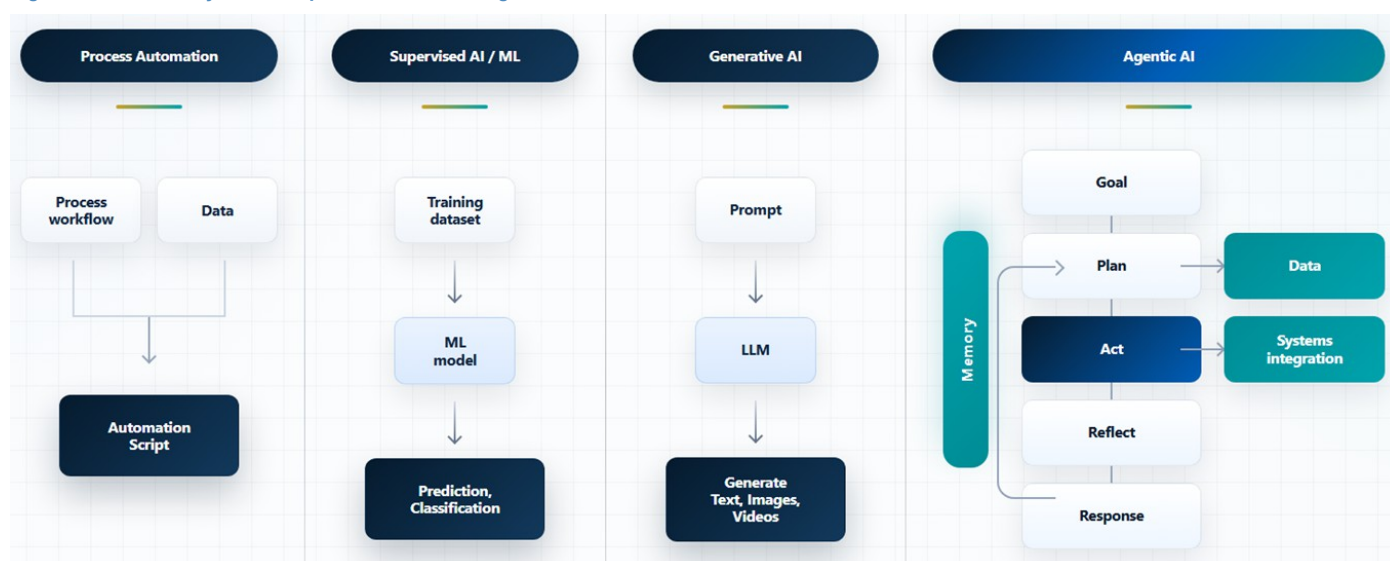
“Reflect” refers to checking intermediate work, identifying errors, and correcting course.

In essence, an agent is not simply a chatbot. It is a system designed to **plan, act, and reflect** in pursuit of an objective. The session was designed to advance the focus from exploring the capabilities of increasingly clever models, to the practicalities of using agentic systems in a real investment workflow. We sense a desire for this shift in client conversations, as questions move from “can the model do it?” to “can we trust the process?”, which means developers must strive to make outputs reviewable, controllable, and defensible in order to be fit for investment decision-making.

This naturally puts governance at the center. If an agent can retrieve data, call tools, and take actions, we need clarity on what it is allowed to do, how it is supervised, what gets reviewed, and what gets escalated. Guardrails in this context are not cosmetic. They are the controls that bound autonomy within a workflow so the system can be productive without becoming a black-box.

The strong turnout was an apt reflection of the increasing interest in agentic AI, particularly in live market contexts where reliability is just as important as performance. We started by mapping the evolution to this point, from simple automation, through machine learning and generative approaches, toward agentic systems that can plan, act, and reflect.

Figure 2: The Journey from Simple Automation to Agentic AI



Source: J.P. Morgan Global Quantitative Strategy

- **Simple automation** sped up data processing
- **Machine Learning** used pattern recognition to bring predictive capabilities
- **GenAI** enabled interaction with broader unstructured data
- **Agentic AI** orchestrates entire workflows by planning, acting, reviewing and iterating until a certain standard is met

Whilst the aim of Agentic AI is an autonomous system, bounding its controls is non-negotiable.

Besides governance, token cost is becoming an increasingly prominent consideration.

Figure 2 illustrates this progression and frames agentic AI as the next step. Traditional process automation relies on scripts that execute fixed instructions, typically to automate repeatable data-processing workflows. Machine learning steps beyond this rigidity; models are trained on datasets to identify statistical patterns, supporting tasks such as prediction and classification. Generative AI extends the interface again by enabling natural-language interaction with unstructured data, making it easier to extract, summarise, and generate content from multi-modal /text-heavy inputs. Agentic AI builds on this by shifting the emphasis from producing a single output to controlling the workflow itself. Rather than following a fixed code path, an agent can break an objective into sub-tasks, select tools, and sequence actions dynamically, while remaining constrained by permissions, review points, and escalation paths.

**In this framing, *autonomy* does not mean unsupervised: it means *freedom of action* inside a governed, reviewable process.**

### Nothing in Life Comes Free: Selective Implementation to Justify Costs

As practitioners delegate increasingly complex tasks to these systems, the benefits are clear, but so are the costs. This is why the economics of tokens versus human time has become almost as prominent a topic as the technology itself. The practical question for most is not whether AI can help, but whether it is economically rational relative to a human, once we account for time-to-completion, rework, error rates, consistency, and oversight.

A useful way to frame this is by task type: AI tends to be advantaged in scalable, repeatable work such as customer service, document classification, and elements of coding and drafting, while humans remain structurally advantaged in high-stakes judgment, strategic, and creative work. The takeaway is that ROI increasingly lives in implementation selectivity, not the model in isolation.

Figure 3: Idealised Research Workflow



Source: J.P. Morgan Quant Strategy, Google Cloud.

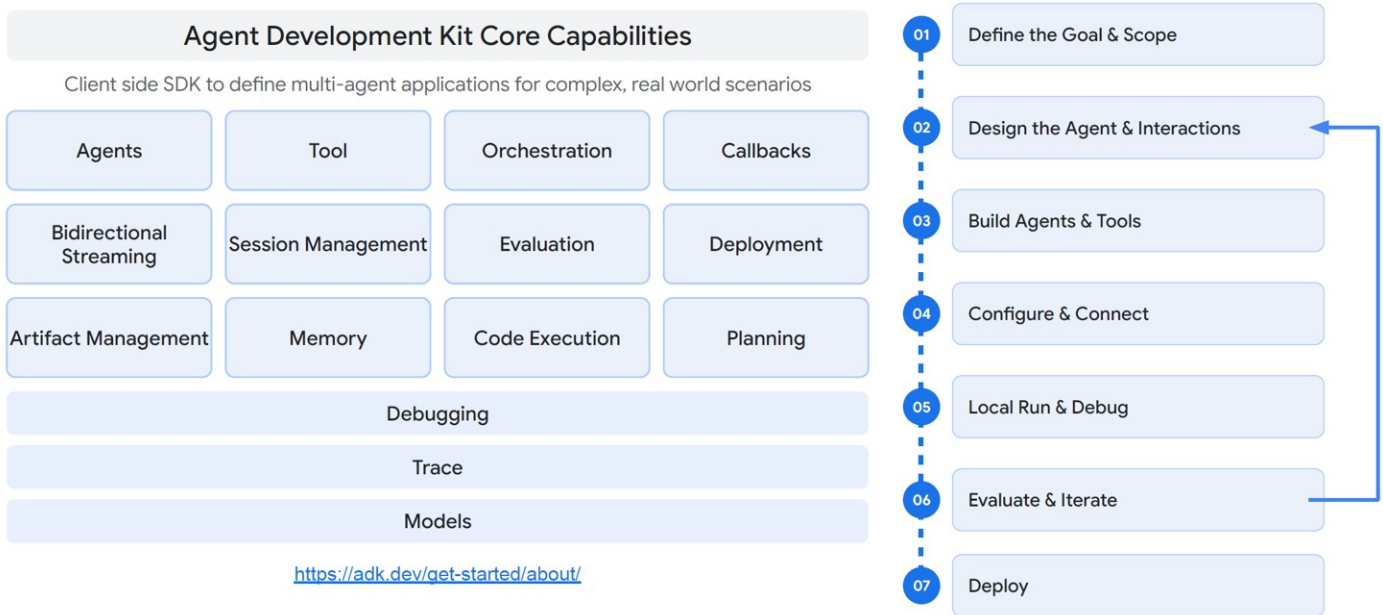
**Open-source SDK:** A collection of publicly accessible programming tools and libraries developers can freely view, modify, and distribute to build applications for a specific platform.

Against that lens, the opportunity in research becomes concrete. Research workflows are naturally multi-step and tool-heavy, which makes them well-suited to agentic automation. But because the output feeds directly into investment decision-making, reliability and traceability are essential, making strict controls a priority. This motivated the case study used throughout the tutorial: financial research agents. Google walked participants through a step-by-step build process, highlighting at each stage the requirements to run it responsibly. The workflow is simple and widely applicable: define the task and constraints, design agent roles and tools including information retrieval and memory, then build, test, and only deploy once the workflow meets reliability standards. We can see the idealised workflow in Figure 3.

### The Workshop Platform: Google Agent Development Kit (ADK)

It has been a remarkable year for agents, and many investors have interacted with them, indirectly at least, through coding assistants and tool-augmented chat workflows, but the challenge is understanding what sits underneath: how instructions are structured, how sub-agents coordinate, and how tools are invoked and governed. The hands-on workshop therefore peeled back the layers using a pro-code approach, starting with simple building blocks and progressing step by step. It was delivered using Google Cloud's Agent Development Kit (ADK), an open-source SDK (software development kit) intended to make it straightforward to build an end-to-end multi-agent system within a short session. Google framed the ADK as a consolidation of the capabilities that teams often spend time piecing together across different interfaces.

Figure 4: Google Agent Development Kit + Workflow



Source: J.P. Morgan Global Quantitative Strategy, Google Cloud.

**Bidirectional Streaming:** real-time interaction where the user can send an input whilst the system responds, and it can stream intermediate results

**Callbacks:** hooks that run at specific points in the agent's execution to intercept and modify behaviour

**MCP Protocol:** a shared standard for how agents call tools

**A2A Protocol:** a shared standard for cross-agent communication across frameworks

**'Tools' are what specialise an agent; they can start as Python functions wrapped for agent use, and later whole specialist agents can be packaged as tools and delegated to by a manager agent.**

- **Platform Capabilities:** the platform emphasises session management, so multi-agent workflows can maintain shared context and memory across a task, and it supports multimodality and bidirectional streaming, so interactions can extend to media beyond text and return partial outputs for review.
- **Standards:** MCP (Model Context Protocol) standardises tool calls and A2A (agent to agent) protocol standardises agent-to-agent communication, making it easier to plug agents into existing systems and to combine specialist components built by different teams without rewriting interfaces.
- **Reviewability, Governance, and Scalability:** callbacks and guardrail hooks let teams set clear limits on which tools an agent can call, insert review steps for higher-risk actions, and define safe fallbacks when outputs are uncertain. Alongside this, agent identity, central registries for agents, tools, and skills, plus gateway and authentication controls support permissioning and oversight at scale. Practical adoption is supported by starter templates in Agent Garden and build-time helpers like the Agent CLI (command line interface), which reduce setup guesswork.

The open-source approach and broad model compatibility helped keep the lessons portable for teams looking to integrate and improve existing workflows, rather than rebuild their stack from scratch.

## Enterprise Agentic AI for Financial Markets

**Firms should be selective to where Agentic AI is implemented, aiming to solve real problems so that gains justify time and monetary costs involved.**

Multi-agent systems are most successful when anchored to a real problem, and then engineered for reviewability at each stage of design, so as to avoid the process transforming into a black-box as complexity is added. Two examples touched on in Google's remarks were a 'horizon' agent that continuously scans government papers across countries to anticipate upcoming regulations, and a retail shopping assistant that uses long term memory so a customer journey can be resumed rather than restarted. Both offer clear commercial benefits, despite markedly distinct objectives. The question for many is what part of their existing workflows would this bring the most benefit to, and it is not a question to be easily dismissed given the time and cost intensity of deploying such systems.

The workshop used a financial research team as the anchor use case, a good fit given that research is naturally multi-step and data-intensive so there are opportunities for automation, but the importance of validity leaves zero room for opacity in reasoning. That said, the capabilities demonstrated are broadly applicable beyond this scope to any workflow where the job is to pull fresh inputs, apply a repeatable analytical process, and produce an output.

**Google showed how to enable agents to access live market data, a tool that nearly all investment workflows are likely to utilise.**

An example, discussed in more detail later in this note, was a tool that lets an agent pull market data at query time. Because an LLM's knowledge is anchored to a training snapshot, many teams still follow a manual pattern: source prices or fundamentals elsewhere, paste them into a prompt, then ask the model to interpret. Tool-enabled agents automate the entire process: retrieve the numbers from an approved source, run the required calculations and then draft the narrative. The headline benefit is speed, which matters in fast-moving markets, but the more important point raised in the workshop was **control**. Restricting permissible data sources, and defining what the agent must do when data is missing, stale, or inconsistent, is essential to avoid confident output built on absent facts.

**When entire workflows become automated, it is easier for hallucinations to propagate through steps undetected.**

Such caution is well placed. **Hallucination** is a real risk, and it becomes harder to detect as systems move from answering questions to running end-to-end workflows, where an early error can propagate through subsequent steps. It is not uncommon for an LLM to produce a plausible answer even when the underlying facts were never retrieved. This is why the session kept returning to governance as a prerequisite for production use, something to be designed in from the start rather than bolted on afterwards.

In practical terms, governance means designing the system so its behavior is reviewable at each step. If a human can't explain how and why an LLM reached its outcome, it should not be trusted. This becomes even more important at scale, where complexity rises and clarity is often the first casualty. The workshop emphasised explicitly defining permissions per agent and per tool, building in review and escalation points with clear acceptance criteria, and setting cost and safety bounds, such as limits on iterations, tool calls, and runtime, to prevent uncontrolled loops and ensure predictable operation.

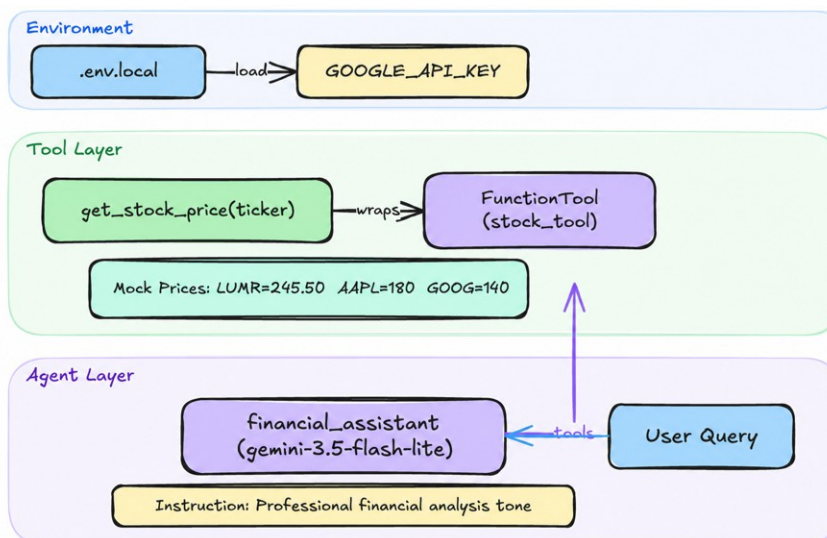
## Specialist Agents for Financial Research

The first *simple* agent created uses a tool to collect stock prices from live financial data sources.

The first module introduced ADK by building a simple 'financial assistant' that can call a single tool to fetch current stock prices.

Crucially, this tool connection bridges the gap between a static model and an evolving market. A standard LLM can only work with what it learned during training plus what you paste into the prompt. With tool access, the agent can pull fresh data and run calculations at query time, so the answer reflects information that did not exist when the model was trained.

Figure 5: End-to-End View - Financial Assistant



Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

Guardrails for when a requested piece of data is not found reduce hallucination risk

Participants launched JupyterLab and configured a Gemini API key so the notebook could authenticate to the selected model. With the environment set, the first step was creating a pricing tool by defining a Python function that takes a ticker and returns a "current" price, defaulting to 100 if the ticker is not found.

Figure 6: Defining the first tool (get\_stock\_price)

```
from typing import Annotated

def get_stock_price(
    ticker: Annotated[str, "The stock ticker symbol (e.g. LUMR, AAPL)"]
) -> float:
    """Fetches the current stock price for a given ticker."""
    print(f"    [Tool] Fetching price for {ticker}...")

    # Simulation Logic (replace with real API in production)
    mock_prices = {
        "LUMR": 245.50,
        "AAPL": 180.00,
        "GOOG": 140.00
    }
    return mock_prices.get(ticker.upper(), 100.00)
```

Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

Next, ADK wraps this function as an invocable tool enabling agent creation. The example below uses the minimal requirements to define an agent - a name, LLM ID, the set of tools it is allowed to utilise, and an instruction.

Figure 7: Creating the agent with the newly-defined tools

The Python function is wrapped as a tool, then imported to the agent's toolkit.

"Instructions" are the prompts which define the agents persona and objective.

```
from google.adk.agents import Agent
from google.adk.tools import FunctionTool

# 1. Wrap the function as an ADK Tool
stock_tool = FunctionTool(get_stock_price)

# 2. Define the Agent
agent = Agent(
    model="gemini-3.5-flash-lite",
    name="financial_assistant",
    tools=[stock_tool], # Give it the tool
    instruction=(
        "You are a helpful Financial Assistant for Lumenridge Financial Group employees. "
        "Use your tools to find stock prices when asked. "
        "Always explain your analysis in a professional tone."
    )
)
```

Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

Now users have a simple agent, the cell below shows how they then interact with it.

Figure 8: The Chat Loop Interface

In-session memory is selected, this is best practice during development

```
from google.adk.runners import Runner
from google.adk.sessions.in_memory_session_service import InMemorySessionService
from google.genai import types
import uuid

# Setup the runner with in-memory storage (good for testing)
runner = Runner(
    agent=agent,
    app_name="module_1_playground",
    session_service=InMemorySessionService(),
    auto_create_session=True
)

# Helper function to run a chat loop
async def chat_loop():
    user_id = "user_1"
    session_id = str(uuid.uuid4())

    print("👋 Financial Assistant Ready! (Type 'quit' to exit)")
    print("-----")

    while True:
        user_input = input("You: ")
        if user_input.lower() in ["quit", "exit"]:
            break

        # Send message to agent
        message = types.Content(role="user", parts=[types.Part(text=user_input)])

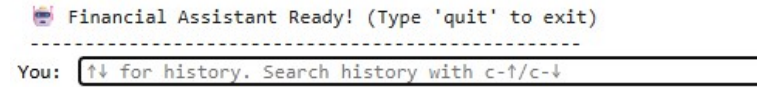
        # Stream the response
        async for event in runner.run_async(
            user_id=user_id,
            session_id=session_id,
            new_message=message
        ):
            if event.content and event.content.parts:
                for part in event.content.parts:
                    if part.text:
                        print(f"Agent: {part.text}")
```

J.P. Morgan Global Quantitative Strategy, Google Cloud

The chat loop appears in the user's notebook after running the code and resembles a conventional AI chatbot

Complex prompts highlight planning ability. The agent recognises it must call the tool multiple times and collate outputs into one answer.

Figure 9: The Chat Loop Interface



Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

The agent takes the returned prices, interprets them in context, and produces a coherent response, rather than simply echoing a number. This enables complex natural language interactions between the user and agent, extending the realm of feasible questions beyond 'What is the price of stock X' to 'Compare the prices of stocks X and Y' or 'Provide an analysis of the current market leaders'.

To validate behavior, the tutorial also uses **ADK Web**, a built-in testing interface that makes interactions observable. Users can inspect each turn's metadata, including whether a tool was invoked, the tool inputs and outputs, and token usage.

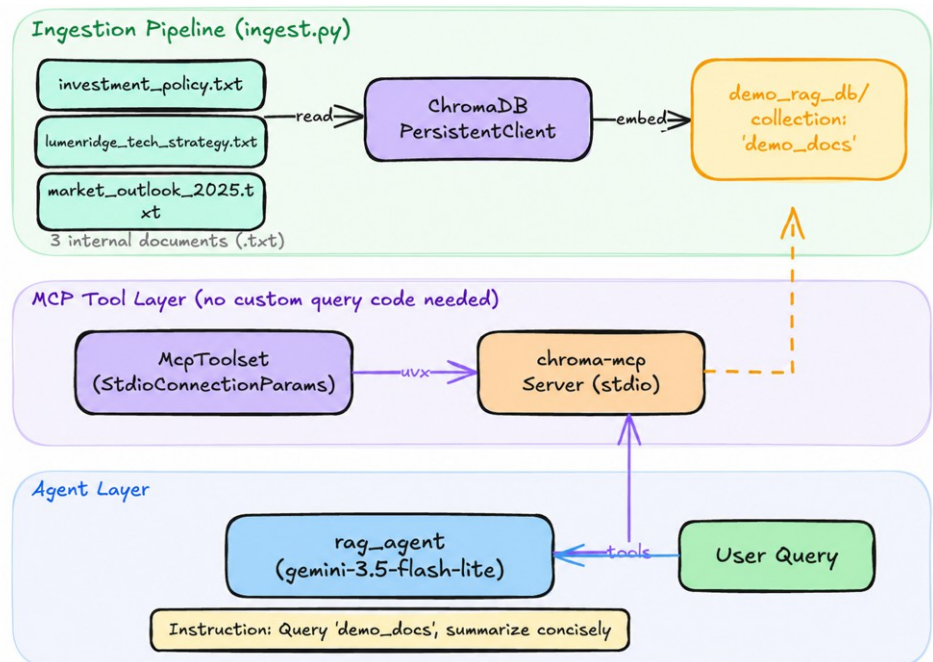
## Specialist Two: the RAG Agent

The second module closely built on the same tool calling pattern but shifted the source of knowledge from live market data to private documents. The goal was to create a RAG (Retrieval Augmented Generation) analyst that can read and reason over private corporate data, such as documents stored on a server or SharePoint, which would be far too large to paste into each prompt.

Documents are split into chunks and converted into embeddings, which are numerical representations of semantic meaning, then these are stored in a vector database.

The user's query is embedded in the same way, and the retriever performs a similarity search to pull back the most relevant passages, which are then provided to the model as grounded context for the answer.

Figure 10: End-to-end view of the RAG analyst



Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

RAG reduces hallucination risk by grounding responses in retrieved excerpts from an approved document set. In the tutorial, the corpus included internal investment policies and market outlooks. In essence, it creates a more firm-specific agent which will stay consistent with house views and relevant constraints.

**Learning the vectorisation process gives developers the freedom to select the agent's knowledge base**

Critically, an understanding of the vectorisation process leaves developers able to select what corpus the agent can “know” at runtime. If the priority is compliance, you vectorise policies, procedures, and regulatory guidance; if the priority is consistency with in-house views, you vectorise published research, outlooks, and strategy notes, with expansive possibilities to tailor this.

The code defines a vector database toolset and reads from a target directory which stores the documents selected for the agent to access. It then defines the RAG agent who is instructed to use this query tool to search the available corpus and return concise, relevant summaries grounded in the retrieved passages. A 30 second connection timeout is added to avoid an agent infinitely scanning for information not contained in its database.

Figure 11: Defining the RAG agent

```
from google.adk.agents import Agent
from google.adk.tools import McpToolset
from google.adk.tools.mcp_tool.mcp_session_manager import StdioConnectionParams, StdioServerParameters
import os
import shutil

# Calculate absolute path so this works from any execution context
db_path = os.path.abspath("./rag_agent/demo_rag_db")

# Define the Tool (uses direct chroma-mcp if installed, or uvx, with a 30s connection timeout)
chroma_cmd = shutil.which("chroma-mcp")
if chroma_cmd:
    mcp_cmd = chroma_cmd
    mcp_args = ["--client-type", "persistent", "--data-dir", db_path]
else:
    mcp_cmd = "uvx"
    mcp_args = ["chroma-mcp", "--client-type", "persistent", "--data-dir", db_path]

chroma_tool = McpToolset(
    connection_params=StdioConnectionParams(
        server_parameters=StdioServerParameters(
            command=mcp_cmd,
            args=mcp_args,
        ),
        timeout=30.0
    )
)

# Create the Agent
# We name it 'root_agent' so it can be picked up by 'adk web'
root_agent = Agent(
    model="gemini-3.5-flash-lite",
    name="rag_agent",
    instruction="""You are a RAG Analyst.
    Use the 'chroma_query_documents' (or similar) tool to answer questions based on the user's document collection.
    The collection name is 'demo_docs'.
    If you find relevant info, summarize it concisely.
    """,
    tools=[chroma_tool]
)

print("✅ Agent defined (in memory)!")
```

Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

**We give the RAG analyst access to the document collection and instruct it on how to use it.**

**The guardrail of a 30 second timeout prohibits an endless loop where an agent may be searching for information not included in its database.**

**An important takeaway is that RAG only becomes reliable with deliberate testing.** Participants were encouraged to run a small set of representative test questions, then inspect the retrieved passages behind each answer to confirm the response is genuinely grounded in the source text. When results look wrong, the diagnostic split is clear: if retrieval is returning the right excerpts but the answer is still poor, tighten the agent instructions and response constraints; if retrieval is missing the right material, optimise the retriever by adjusting chunking, embeddings, and search settings. **In production, this has become a critical discipline; regular spot checks and periodic evaluations preserve trustworthiness as the document set accumulates.**

**Effective RAG implementation prioritises testing to verify the model is quoting the intended evidence.**

## Manager-Led Orchestration

**Routing adds both quality and cost control.**

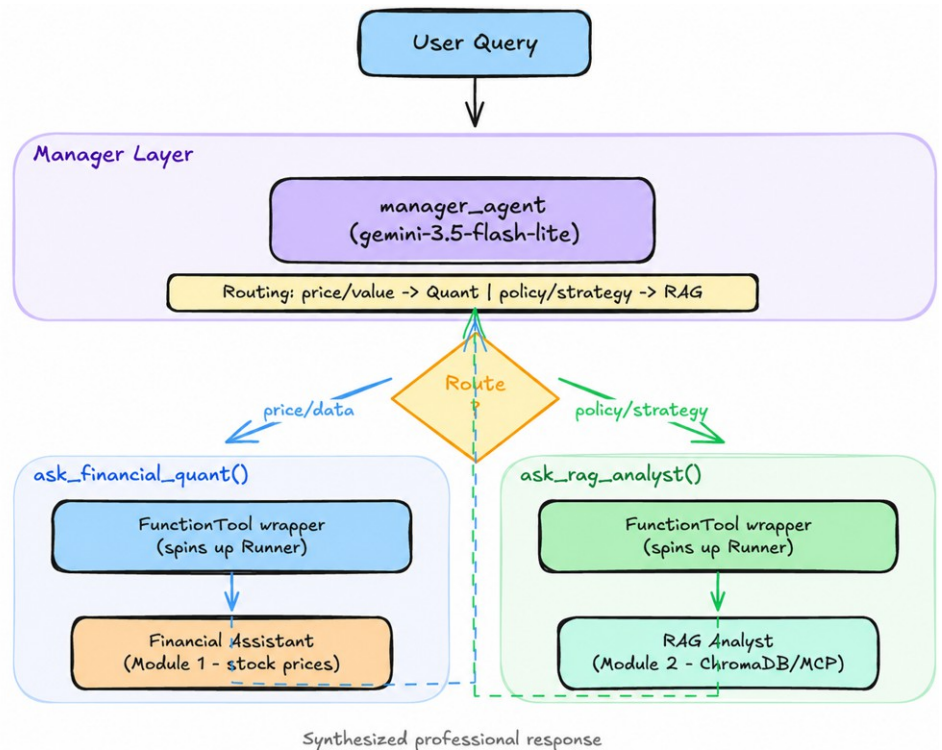
**A routing agent increases output quality as there is less room for confusion when agents focus on a tight remit.**

**It also cuts costs by directing routine sub-tasks to cheaper, faster models and reserving higher-capability models for genuinely complex reasoning.**

Module 3 builds a hierarchical workflow agent capable of delegating tasks and managing multi-step reasoning processes. Many call this a routing agent, here we call it the manager.

Giving one agent too many roles or tasks can confuse it and decrease the accuracy of the output. It is also economically inefficient as cost minimisation typically means reserving frontier models for genuinely complex reasoning, whilst routine tasks can be run on faster, cheaper models. This is where the manager adds value. Rather than doing the analysis itself, it identifies the user's objective, plans a series of steps to achieve it, then routes each to the most suitable specialist worker agent, before collating the outputs into a single, coherent response.

Figure 12: End-to-End View - the Manager Agent



Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

The two analysts constructed in the earlier modules - the Financial Analyst and the RAG Agent - are imported. In this way each agent becomes a tool to the manager.

The interface's focus on **modularity** adds benefits here for organisations as teams can construct specialised agents, package them as tools and share them, allowing for various multi-agent architectures to be created.

Figure 13: Importing the team - the Financial Market Analyst and the RAG Analyst

```
try:
    from module_01.financial_agent_app.agent import root_agent as financial_agent
    from module_02.rag_agent.agent import root_agent as rag_agent
    print("✅ Team recruited: Financial Agent and RAG Agent are ready.")
except ImportError as e:
    print(f"❌ Import Error: {e}")
    print("Make sure you have completed Module 1 and 2 and they have 'agent.py' files.")
```

Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

This shifts a tool from being a function to an entire agent.

Instead of the manager calling an API like `get_stock_price`, it calls a specialist agent that already knows how to use this tool, which is more lightweight and cost effective.

The notebook wrapped each specialist agent inside a Python function so it could be exposed to the manager as a callable tool.

Figure 14: Defining the Manager's Toolkit

```
from typing import Annotated
from google.adk.runners import Runner
from google.adk.sessions.in_memory_session_service import InMemorySessionService
from google.genai import types

async def ask_financial_quant(
    question: Annotated[str, "The financial question to ask (e.g. stock price, ROI)"]
) -> str:
    """Delegates a question to the Financial Quant agent."""
    print(f" [Manager] Delegating to Financial Agent: '{question}'")

    runner = Runner(
        agent=financial_agent,
        app_name="financial_sub_task",
        session_service=InMemorySessionService(),
        auto_create_session=True
    )

    response_text = ""
    # We iterate through the response to get the final answer
    async for event in runner.run_async(
        user_id="manager",
        session_id="session_sub_1",
        new_message=types.Content(role="user", parts=[types.Part(text=question)])
    ):
        if event.content and event.content.parts:
            for part in event.content.parts:
                if part.text:
                    response_text += part.text

    return response_text

async def ask_rag_analyst(
    question: Annotated[str, "The document-based question to ask (policies, strategy)"]
) -> str:
    """Delegates a question to the RAG Analyst agent."""
    print(f" [Manager] Delegating to RAG Agent: '{question}'")

    runner = Runner(
        agent=rag_agent,
        app_name="rag_sub_task",
        session_service=InMemorySessionService(),
        auto_create_session=True
    )

    response_text = ""
    async for event in runner.run_async(
        user_id="manager",
        session_id="session_sub_2",
        new_message=types.Content(role="user", parts=[types.Part(text=question)])
    ):
        if event.content and event.content.parts:
            for part in event.content.parts:
                if part.text:
                    response_text += part.text

    return response_text
```

Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

Next the manager is defined. Its primary function is routing, but it also exists to collate the outputs of multiple agents, and is able to track the progress of a multistep inquiry, ensuring all subtasks are completed before it responds to the user.

Figure 15: Creating the Manager Agent

```
from google.adk.agents import Agent
from google.adk.tools import FunctionTool

manager_agent = Agent(
    model="gemini-3.5-flash-lite",
    name="manager_agent",
    tools=[
        FunctionTool(ask_financial_quant),
        FunctionTool(ask_rag_analyst)
    ],
    instruction=(
        "You are a Senior Investment Manager. \n"
        "You have a team of experts:\n"
        "1. Financial Quant: Ask for real-time data like stock prices.\n"
        "2. RAG Analyst: Ask for internal knowledge like policies and strategy.\n\n"
        "Routing Rules:\n"
        "- If the user asks about 'price', 'value', or 'market data', ask the Financial Quant.\n"
        "- If the user asks about 'policy', 'strategy', 'guidelines', or 'outlook', ask the RAG Analyst.\n"
        "- Always synthesize the answer and present it professionally."
    )
)

print("✅ Manager Agent is ready to rule!")
```

Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

**The benefit of a routing agent comes from scale, when the quantity of tools and sub-agents grows beyond a humans ability to keep track of them.**

**Testing is again the main piece of advice for proficient implementation.**

In this module, a question like “What is the price of stock X?” should return the same result as in the earlier specialist-agent setup, but instead of the user selecting the financial analyst agent explicitly, a manager agent infers intent and routes the request to the right specialist and tools automatically. In this basic example, it can feel like an unnecessary step. The value shows up when the system grows to dozens or hundreds of specialists with distinct capabilities. At that point, routing becomes a cognitive bottleneck for humans, who cannot reliably remember every agent’s remit and tool access, whereas an agent does not hold the same limitation.

For Production, there was a repeated emphasis on experimentation. Switching the LLM is lightweight and typically requires changing a single line of code, so teams were encouraged to test which tasks can be handled by cheaper, faster models versus which genuinely require higher capability models. They can then tune the routing rules around those findings, balancing accuracy, latency, and cost. It is not a one size fits all process, and the biggest benefit comes from tailoring the configuration to the specific workflow and requirements.

## Complex Multi-Agent Systems in Practice

Controls must be thoughtfully ingrained in design. They specify which tools or subagents can run, when they run, and what “good enough” looks like, rather than letting an unconstrained agent loop indefinitely.

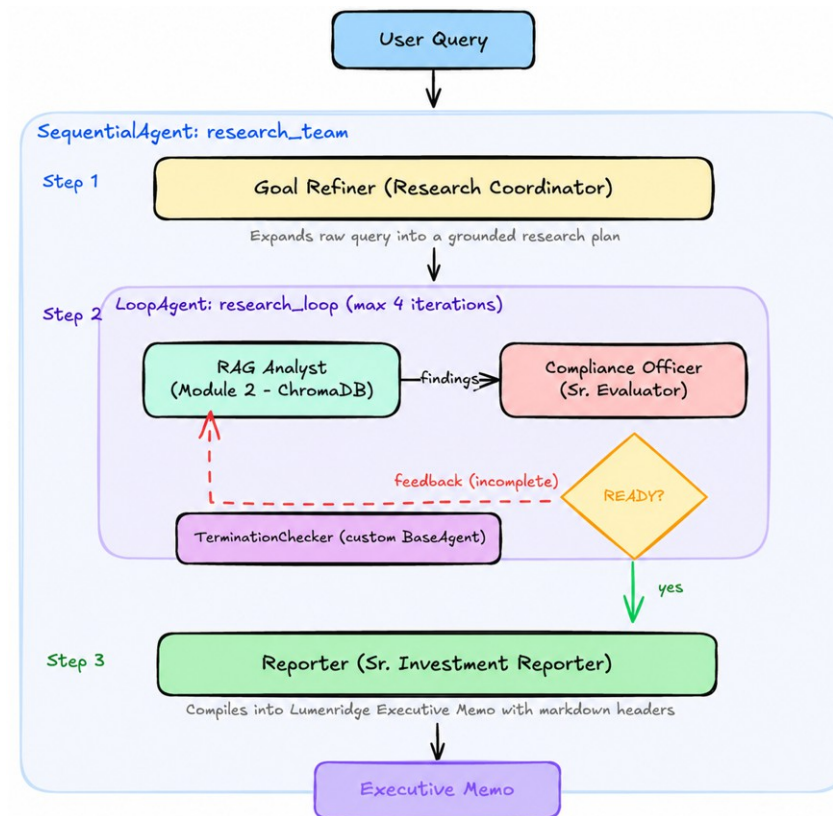
For instance, a maximum iteration count can be implemented so that if the evidence is still insufficient after the cap, the system can fail fast and return to the user, rather than consuming tokens for marginal improvement.

The final module collated all prior building blocks to form an autonomous research team. The framing was intentionally familiar. In a small team, one analyst can often take a question end-to-end. At scale, a manager is needed to break tasks into workstreams, enforce standards, and own the final call. The agentic design followed the same logic. Rather than relying on a single agent to do everything, it used hierarchical orchestration to split a complex request into clear phases, with explicit acceptance criteria and loop limits to control both quality and cost.

This is implemented as a sequential three-stage pipeline:

- The goal refiner:** a planner agent turns a high-level prompt into a precise research brief. It is instructed to plan, not answer, and to produce clear instructions for downstream specialists
- The research loop:** a RAG analyst retrieves supporting evidence from the document base, a compliance officer reviews the draft findings, and a termination checker decides whether the work is ready or needs another iteration.
- The reporter:** a writer agent synthesises a polished executive memo using only the research loop output, strictly instructed not to fill gaps with its own knowledge.

Figure 16: End-to-End view - Autonomous Research Team



Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

As in earlier modules, the first step of the notebook imports the financial analyst and the RAG agent as tools.

Next, the notebook defines the research team roles and their instructions, with the compliance officer acting as an explicit quality gate and the reporter constrained to use only retrieved, verified claims.

**Agent instructions are a key lever to bound their autonomy - for example, we tell the reporter to 'never add outside facts' which controls data sourcing.**

Figure 17: Defining the Research Team

```
# A. Goal Refiner
goal_refiner = Agent(
    model="gemini-3.5-flash-lite",
    name="goal_refiner",
    instruction="""
You are the Research Coordinator for Lumenridge Financial Group. Turn the user's request into a
research plan; do not answer it yourself. Tell the analyst to use only facts retrieved from the
supplied workshop documents and never supplement them with general knowledge. For a strategy memo,
require evidence on the AI-infrastructure thesis, its valuation and growth thresholds, the Clean
Future ESG rules, and the portfolio risk controls. Request relevant internal technology controls
when the question calls for them. If a requested fact is absent, tell the analyst to say so. Do not
introduce unverified companies, figures, regulations, or technologies into the plan.
"""
)

# B. Compliance Officer (The Loop Critic)
compliance_officer = Agent(
    model="gemini-3.5-flash-lite",
    name="compliance_officer",
    instruction="""
You are the Senior Compliance Officer for Lumenridge Financial Group. Treat retrieved workshop-
document content as the only valid evidence. Review every material claim and reject anything not
supported by that evidence. Never accept or add outside facts, regulations, technologies, companies,
or figures; GDPR, Basel III, the EU AI Act, Kubernetes, and PII controls are unsupported unless they
were actually retrieved. For a strategy memo, require the Lumenridge name and supported facts on the
AI-infrastructure thesis, valuation and growth thresholds, Clean Future ESG rules, and portfolio risk
controls, plus technology facts requested by the user. If evidence is missing or a claim is unsupported,
direct the analyst to retrieve evidence, remove the claim, or say that the documents do not specify it.
Only when the findings are complete, professional, and fully grounded, start your response with:
READY_FOR_SUMMARY.
"""
    ,
    output_key="compliance_report" # Important: Save output so the Checker can read it
)

# C. Reporter (The Endpoint)
reporter = Agent(
    model="gemini-3.5-flash-lite",
    name="reporter",
    instruction="""
You are the Senior Investment Reporter for Lumenridge Financial Group. Write a concise Markdown
executive memo using only claims from the analyst's retrieved workshop-document evidence that the
compliance review accepted. Compliance feedback and earlier agent prose are not evidence. Brand the
memo only as Lumenridge Financial Group. Include supported findings on the AI-infrastructure thesis,
valuation and growth thresholds, Clean Future ESG rules, and portfolio risk controls, plus relevant
internal technology facts requested by the user. Never add outside facts, regulations, technologies,
companies, or figures. In particular, omit GDPR, Basel III, the EU AI Act, Kubernetes, and PII controls
unless retrieved. Do not fill gaps with plausible details; omit the claim or say it is not specified
in the workshop documents. If compliance did not emit READY_FOR_SUMMARY, return a short notice that
the memo was not approved instead of presenting unverified findings. Use clear headings and make
recommendations direct syntheses of sourced
policy, not new financial advice.
"""
)

print("✅ Orchestration components ready.")
```

Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

Finally, with a well-defined team, the agents are assembled into composite structures: a sequential agent for the end to end pipeline, plus a loop agent for iterative refinement and self-correction.

Grouping the agents dictates how the cross-agent interactions occur.

In this case, a goal refiner turns the user prompt into a clear brief which feeds into a three agent loop. These agents iterate until their output meets the acceptance bar or hits a limit, at which stage output passes to a reporter who produces the final memo.

Figure 18: Building the Composite Agents

```
# 1. The custom Termination Checker
# This agent acts as a logic gate inside the loop.
# It reads the "compliance_report" from the session state.
class TerminationChecker(BaseAgent):
    async def _run_async_impl(self, ctx):
        # Read the previous agent's output from state
        report = ctx.session.state.get("compliance_report", "")

        if report.lstrip().startswith("READY_FOR_SUMMARY"):
            # Signal the Loop to stop immediately
            yield Event(
                author=self.name,
                actions=EventActions(escalate=True)
            )
        else:
            # Continue the loop
            yield Event(author=self.name)

termination_checker = TerminationChecker(name="termination_checker")

# 2. The Autonomous Loop
research_loop = LoopAgent(
    name="research_loop",
    sub_agents=[rag_analyst, compliance_officer, termination_checker],
    max_iterations=4
)

# 3. The Full Research Team Pipeline
research_team = SequentialAgent(
    name="research_team",
    sub_agents=[goal_refiner, research_loop, reporter]
)

print("🚀 Research Team is assembled.")
```

Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

Once all of the components have been defined and imported, the user interface is the same chat bot from the first module:

Figure 19: In-Notebook Agent Chat Loop

```
🤖 Research Team Ready! (Type 'quit' to exit)
-----
User (type 'quit' to exit): 
```

Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

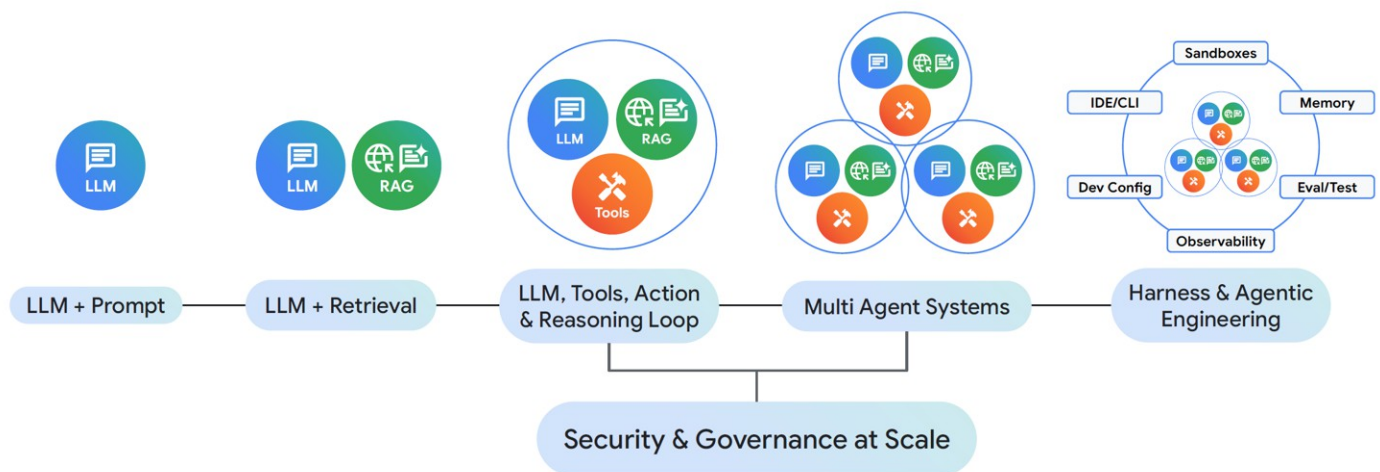
**ADK Web enables bi-directional streaming which makes it easier to diagnose issues.**

As users delegate increasingly complex tasks to these systems, the ADK's bi-directional streaming capabilities grow more beneficial. ADK Web allows users to check the progress of their calls by requesting intermediary outputs, allowing visibility on where issues may stem from, and allowing the necessary refinements to be targeted at pain points.

## Autonomous Financial Research

The closing demo and Q&A framed around an evolution that is easy to recognise in practice, from a prompt-only LLM, to retrieval grounding, to tool-using reasoning loops, to multi-agent teams, and finally to the “harness” layer that makes these systems operable at scale.

Figure 20: Agentic AI - An Evolution: What is next?



Source: J.P. Morgan Global Quantitative Strategy, Google Cloud

**A harness is the difference between a clever prototype and a deployable system.**

**It exposes what is happening behind the scenes as it happens - not just a final outcome.**

**Agentic performance is built, not bought off the shelf.**

**Test your own workflow - vary the model choice, inference settings, instructions and tool set until the system behaves consistently ...**

**...and a message becoming increasingly topical: Governance is not an afterthought!**

A harness is the software layer that wraps around AI models and manages the execution environment, so the system can load the right skills and tools, generate or run code where needed, and coordinate agent-to-agent interactions to complete longer tasks end-to-end. Crucially, it keeps the workflow reviewable by exposing the plan and step-by-step progress, including which sub-agents and skills were used, so users can intervene when necessary and prevent the process becoming a black-box.

The practical demonstration of a multi-agent system with a real use-case in financial research left most participants thinking about how to translate the pattern into their own environments. When making such considerations, two takeaways stood out:

- **Testing and workflow-specific tuning matter; there is no one-size-fits-all configuration.** Teams were encouraged to iterate to make the system best fit their needs. Three practical levers were highlighted: **1)** inference parameters such as temperature to reduce variability, **2)** richer system instructions including few-shot examples to standardise outputs, and **3)** testing to decide where to split tools versus delegate to sub-agents
- **Governance must be engineered, not assumed.** A production harness should make the workflow inspectable, including step-by-step actions, tool invocations, and resource usage, so outputs can be reviewed, tuned, and defended as systems scale. Building such systems from the ground up gives developers a concrete understanding of the component parts, which reduces the chances of the reasoning process becoming a black-box.

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